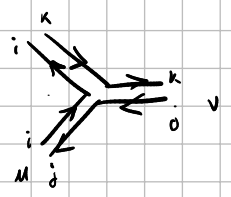
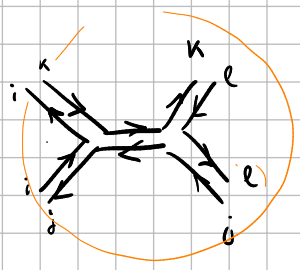
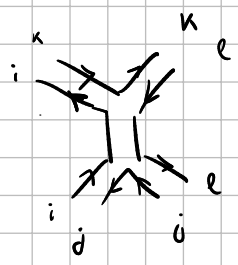


$(A_\mu)^i{}_j$



s channel

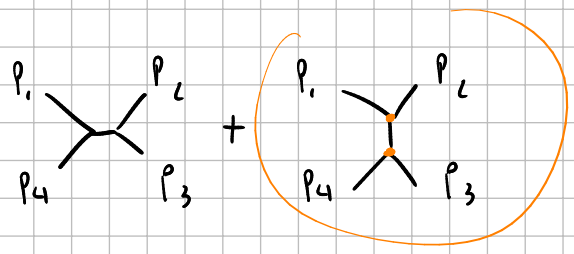


t channel

planar graphs

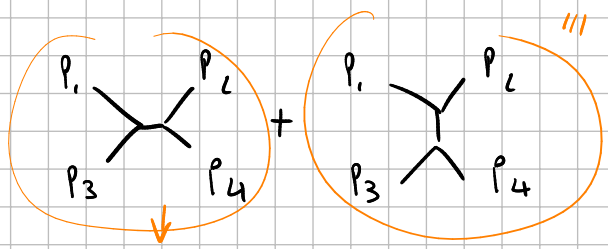
$$\mathcal{A}_n = \sum_{\sigma \in S_n/\mathbb{Z}_n} (- \text{—}) \, A_n(\sigma(1), \sigma(2), \dots, \sigma(n))$$

$A_4(1,2,3,4) =$

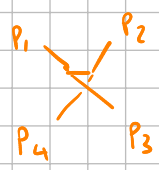


$\to \dfrac{1}{(p_1+p_2)^2}$

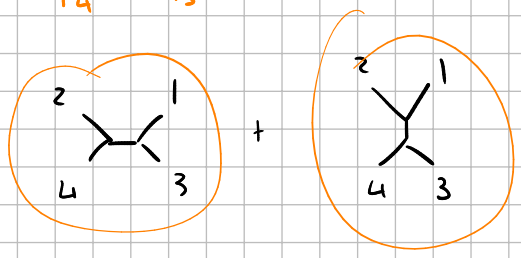
$A_4(1,2,4,3) =$

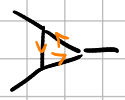


$\to \dfrac{1}{(p_1+p_2)^2}$

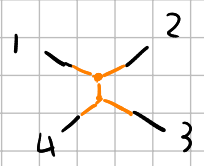


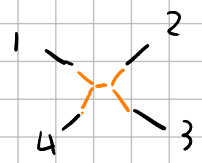
$A_4(21\ 34) =$



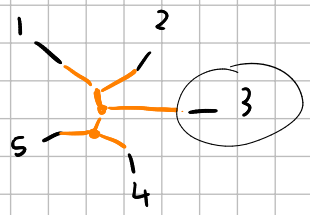
- tree-level diagrams: no loops  not allowed!
- fixed ordering
- planar diagrams
- connected

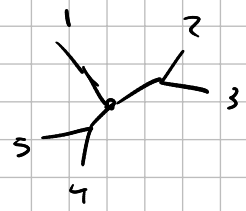
$n = 4$

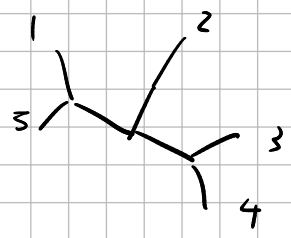




$n = 5$

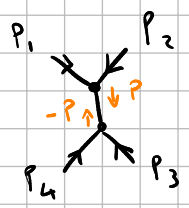




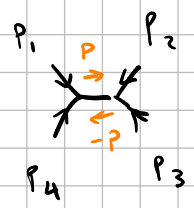


<u>rule</u> each internal edge is a propagator $\frac{1}{p^2}$

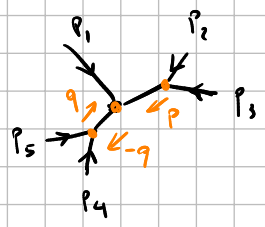
at each vertex we have conservation of momentum



$$\frac{1}{p^2} = \frac{1}{(p_1+p_2)^2} = \frac{1}{(-p_3-p_4)^2}$$

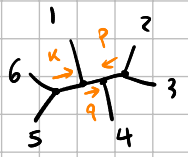


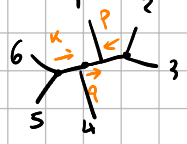
$$\frac{1}{p^2} = \frac{1}{(p_1+p_4)^2} = \frac{1}{(p_2+p_3)^2}$$



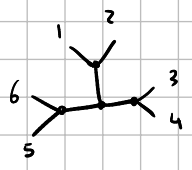
$$\frac{1}{(p_2+p_3)^2} \quad \frac{1}{(p_4+p_5)^2}$$

$$-q = p_1 + p = p_1 + p_2 + p_3 = -p_4 - p_5$$



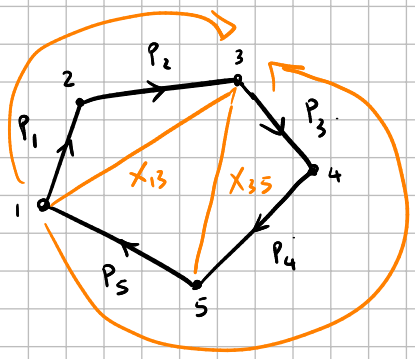


$$\underset{X_{24}}{\frac{1}{(p_2+p_3)^2}} \cdot \underset{X_{25}}{\frac{1}{(p_2+p_3+p_4)^2}} \cdot \underset{X_{51}}{\frac{1}{(p_5+p_6)^2}}$$



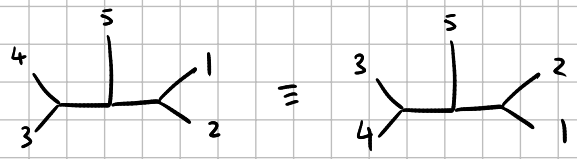
$$\frac{1}{(p_1+p_2)^2}\cdot\frac{1}{(p_3+p_4)^2}\cdot\frac{1}{(p_5+p_6)^2}$$

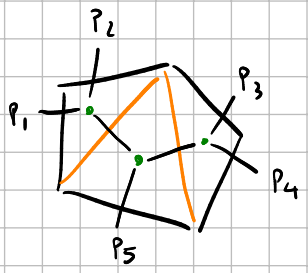
x_{13} $\quad$ x_{35} $\quad$ x_{51}



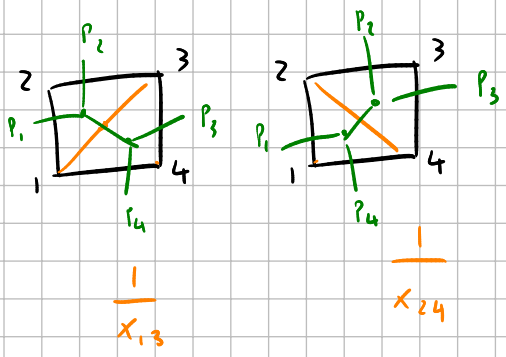
$$\sum_i p_i = 0$$

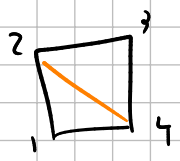
$$\frac{1}{x_{13}}\,\frac{1}{x_{35}} = \frac{1}{(p_1+p_2)^2}\,\frac{1}{(p_3+p_4)^2}$$

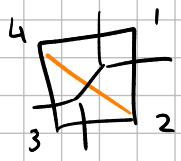




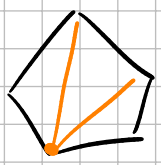
$n=4$





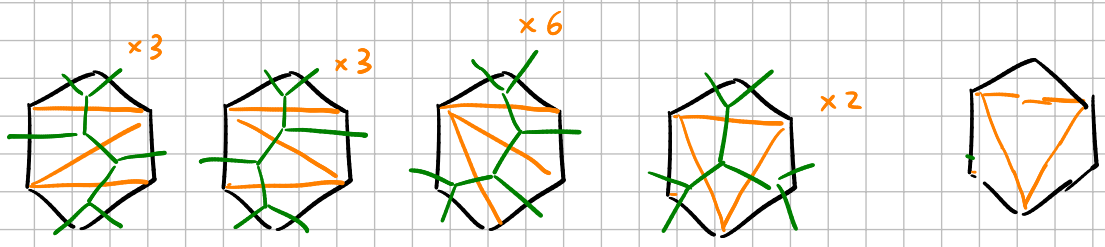


$n=5$



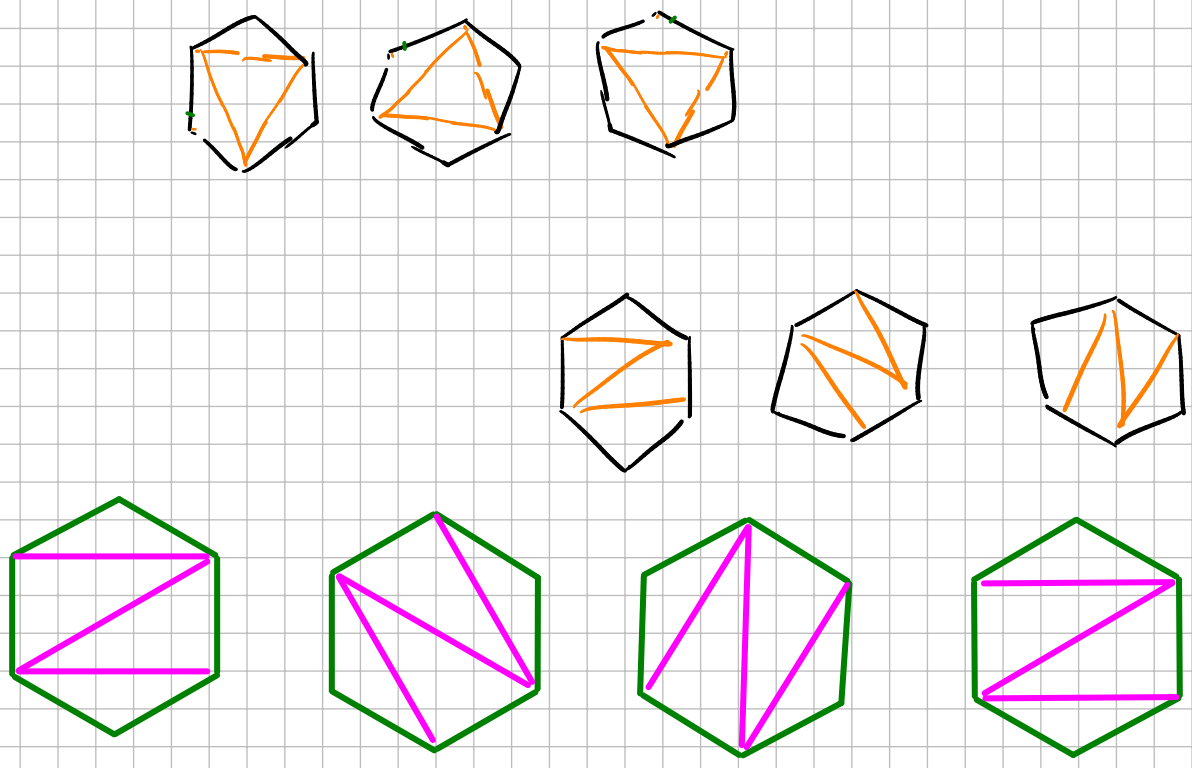
+ cyclic

$n=6$



number of planar diagrams with fixed ordering is the Catalan number

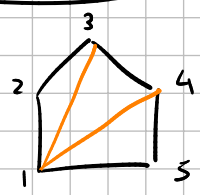
C_k $\quad C_0 = 1 \quad C_1 = 1 \quad C_2 = 2 \quad C_3 = 5 \quad C_4 = 14$



$$\frac{1}{n+1}\binom{2n}{n}$$

---

triangulation τ



pick X_{13} X_{14} as planar

everything but $c_{i-1,j-1}$ with $(i,j) \in \{(1,3),(1,4)\}$

exclude c_{52} c_{53} and c_{15} c_{45} → because planar

and c_{12} c_{23} c_{34}

$n = 5$

number of invariants: $\frac{n(n-3)}{2} = 5$

c_{12} c_{13} c_{14} $\cancel{c_{15}}$

c_{23} c_{24} $\cancel{c_{25}}$

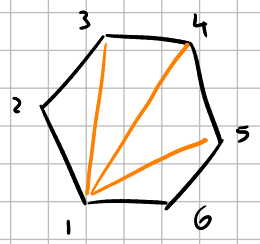
c_{34} $\cancel{c_{35}}$

$\cancel{c_{45}}$

$$c_{12} = -X_{13} = -(P_1+P_2)^2$$

planar: sum of consecutive momenta squared

$$c_{ij} = -(P_i+P_j)^2$$



take X_{13} X_{14} X_{15} | exclude c_{62} c_{63} c_{64}

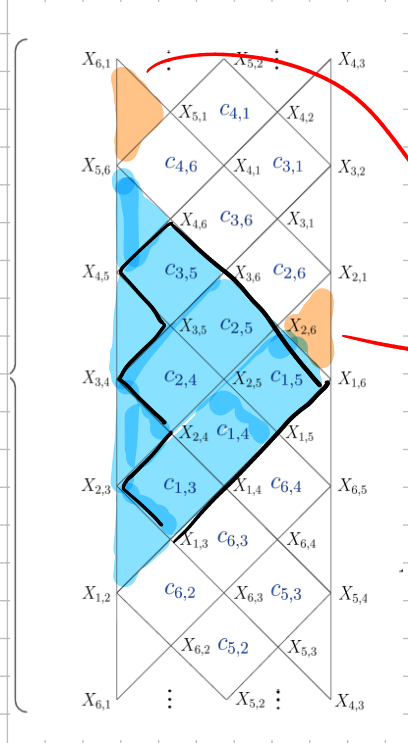
$c_{i-1,j-1}$

c_{12}	c_{13}	c_{14}	c_{15}	c_{16}
	c_{23}	c_{24}	c_{25}	$\cancel{c_{26}}$
		c_{34}	c_{35}	$\cancel{c_{36}}$
			c_{45}	$\cancel{c_{46}}$
				c_{56}

$c_{ij} = -(p_i+p_j)^2$

replace

X_{24}	with	c_{13}
X_{25}	with	c_{24}
X_{26}	with	c_{15}
X_{35}		c_{24}
X_{36}		c_{25}
X_{46}		c_{35}



planar $c_{16} = -X_{62}$

$n=6$: 9 independent variables

---

in the $n=5$ example:

X_{13} X_{14} $\cancel{c_{12}}$ c_{13} c_{14} $\cancel{c_{23}}$ c_{24} $\cancel{c_{34}}$

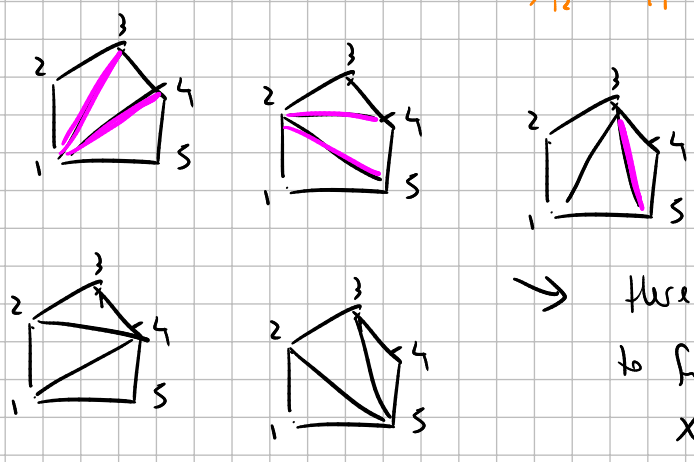
=

non planar

want to show that the remaining ones (X_{24} X_{25} X_{35}) are obtained from basis

X_{13} X_{14}

$X_{12} = p_1^2 = 0$



→ these pictures are to find the non-zero X_{ij} s

$$c_{ij} = X_{ij} + X_{i+1\,j+1} - X_{i\,j+1} - X_{i+1\,j}$$

$$c_{13} = X_{13} + X_{24} - X_{14}$$

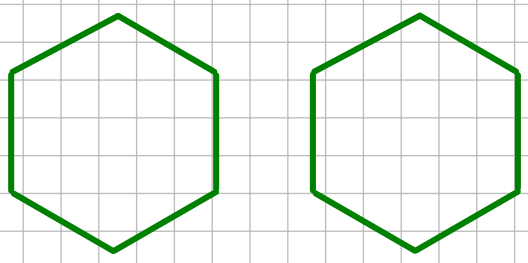
$$c_{14} = X_{14} + X_{25} - X_{24}$$

$$c_{24} = X_{24} + X_{35} - X_{25}$$

$$X_{25} = c_{14} + c_{13} - X_{13}$$

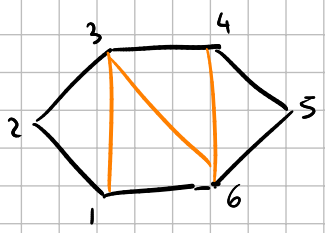
$$X_{35} = c_{14} + c_{24} - X_{14}$$

$$X_{24} = c_{24} - X_{35} + X_{25}$$



<u>Procedure</u> to replace variables:

- choose triangulation, keep the chords
- for each discarded X_{ij}, replace with $c_{i-1,\,j-1}$



X_{13} X_{36} X_{46}

<u>replace:</u>

$X_{14} \to c_{63}$

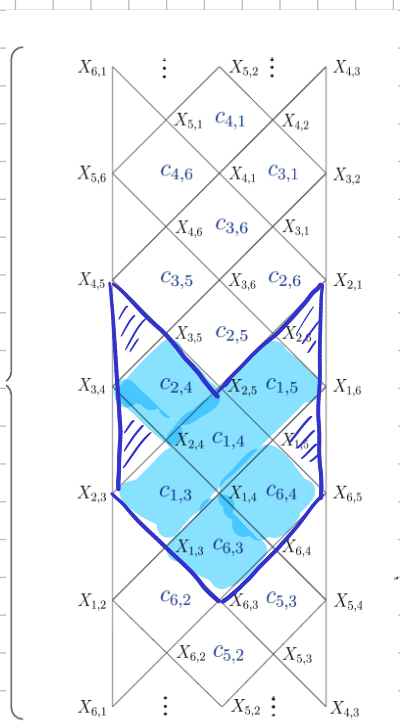
$X_{15} \to c_{64}$

$X_{24} \to c_{13}$

$X_{25} \to c_{14}$

$X_{26} \to c_{15}$

$X_{35} \to c_{24}$



$$c_{23} = \overset{0}{X_{23}} + \overset{0}{X_{34}} - X_{24} - \overset{0}{X_{33}}$$

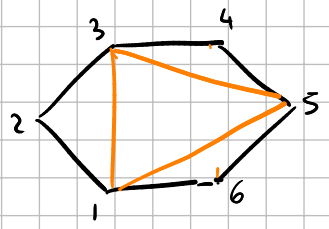
$$\parallel$$

$$-X_{24}$$

if $i \leq j$ $\quad X_{ij} = \left(\sum_{k=i}^{j-1} p_k\right)^2$

$X_{i\,i+1} = p_i^2 = 0$

$X_{ii} = 0$ by definition



X_{13} X_{35} X_{51}

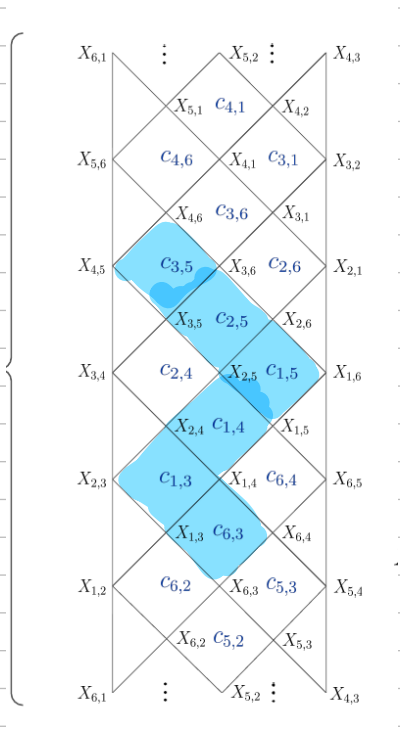
<u>replace:</u>

$X_{14} \to C_{63}$

$X_{24} \to C_{13}$

$X_{25} \to C_{14}$

$X_{26} \to C_{15}$

$X_{36} \to C_{25}$

$X_{46} \to C_{35}$